

ORDER DATA ANALYSIS & FORECASTING



channelengine

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OUTLINE

1. Introduction
2. Data cleaning
3. Data analysis
4. Forecasting
5. Conclusions
6. Q&A



OUTLINE

1. Introduction

1.1 Datasets

1.2 Business significance

1.3 Research questions

1.4 Methods

2. Data cleaning

3. Data analysis

4. Forecasting

5. Conclusions

6. Q&A



1. INTRODUCTION

1.1 Datasets

channel	2/8/2022 6:59 PM	JSON File	3 KB
company	2/8/2022 4:51 PM	JSON File	1 KB
order_status	2/8/2022 9:01 PM	JSON File	1 KB
tenant	2/8/2022 4:51 PM	JSON File	2 KB
tenant_Acme_Corp_APAC	2/8/2022 7:50 PM	OpenOffice.org 1....	207 KB
tenant_Acme_Corp_EMEA	2/8/2022 7:53 PM	OpenOffice.org 1....	118 KB
tenant_Acme_Corp_US	2/8/2022 7:48 PM	OpenOffice.org 1....	532 KB
tenant_BuYaYa BV	2/8/2022 7:43 PM	OpenOffice.org 1....	3,942 KB
tenant_Green_is_good_BV	2/8/2022 7:40 PM	OpenOffice.org 1....	9,432 KB
tenant_Horizon_Zone_APAC	2/8/2022 7:47 PM	OpenOffice.org 1....	502 KB
tenant_Horizon_Zone_CA	2/8/2022 7:49 PM	OpenOffice.org 1....	470 KB
tenant_Horizon_Zone_EMEA	2/8/2022 7:45 PM	OpenOffice.org 1....	1,176 KB
tenant_Horizon_Zone_US	2/8/2022 7:51 PM	OpenOffice.org 1....	126 KB
tenant_Jelly_b2c	2/8/2022 7:44 PM	OpenOffice.org 1....	3,227 KB
tenant_Sofa_team_BV	2/8/2022 7:54 PM	OpenOffice.org 1....	40 KB
tenant_Bitterballen_BV.parq	2/8/2022 7:27 PM	PARQ File	13,803 KB
tenant_Horizon_Zone_Global.parq	2/8/2022 7:20 PM	PARQ File	77 KB
tenant_Jelly_b2b.parq	2/8/2022 7:35 PM	PARQ File	21,289 KB

- **4 json files:** dimension (company, tenant, channel, order status)
- **11 csv + 3 parq files:** orders data.
 - 14 columns
 - over 1.4 million records

	order_date	company_id	tenant_id	order_id	order_line_id	p
0	2022-01-19	392	7406	4155075457	61550753855797	
1	2022-01-18	392	7406	9285965277	42859653855886	
2	2022-01-18	392	7406	5176965060	9176965385511	
3	2022-01-19	392	7406	3858075698	58580753855749	
4	2022-01-19	392	7406	3068075412	9068075385575	
...	...	...	...	...	...	
429	2021-07-11	973	4851	5930	293855971	
430	2022-01-27	973	4851	4373831	23733855219	
431	2022-02-02	973	4851	2504458	95043855416	
432	2022-01-30	973	4851	259315	15933855148	
433	2022-01-29	973	4851	9683890	8683385554	

1428652 rows × 14 columns

1. INTRODUCTION

1.2 Business significance

Past performance

- Temporal trend
- Channel performance
- Category performance
- Order status
- Price



Forecasting

- Plan inventory
- Plan manpower
- Future investment



1. INTRODUCTION

1.3 Research questions

- **How can we measure the past performance for our clients?**
 - How does performance fluctuate over time?
 - What are the best performing channels?
 - What are the best performing product categories?
 - What does the distribution of order status look like?
 - How does price change over time and does it affect performance?
- **How can we forecast future trends for our clients?**
 - How can ARIMA model be used to forecast future order quantity?
 - How can we find the best parameters for ARIMA model?
 - How can we implement seasonality in ARIMA model?
 - What are the limitations and how can we improve our forecasting model in the future?



1. INTRODUCTION

1.4 Methods

Data quality check

- Completeness
- Uniqueness
- Consistency
- Accuracy

Data analysis

- Visual analysis
 - Temporal change
 - Channel/Category deepdive
 - Order status
 - Price

Forecasting

- ARIMA
 - Non-seasonal
 - Seasonal

OUTLINE

1. Introduction

2. Data cleaning

2.1 Data pipeline

2.2 Data quality issues

2.3 Data correction

3. Data analysis

4. Forecasting

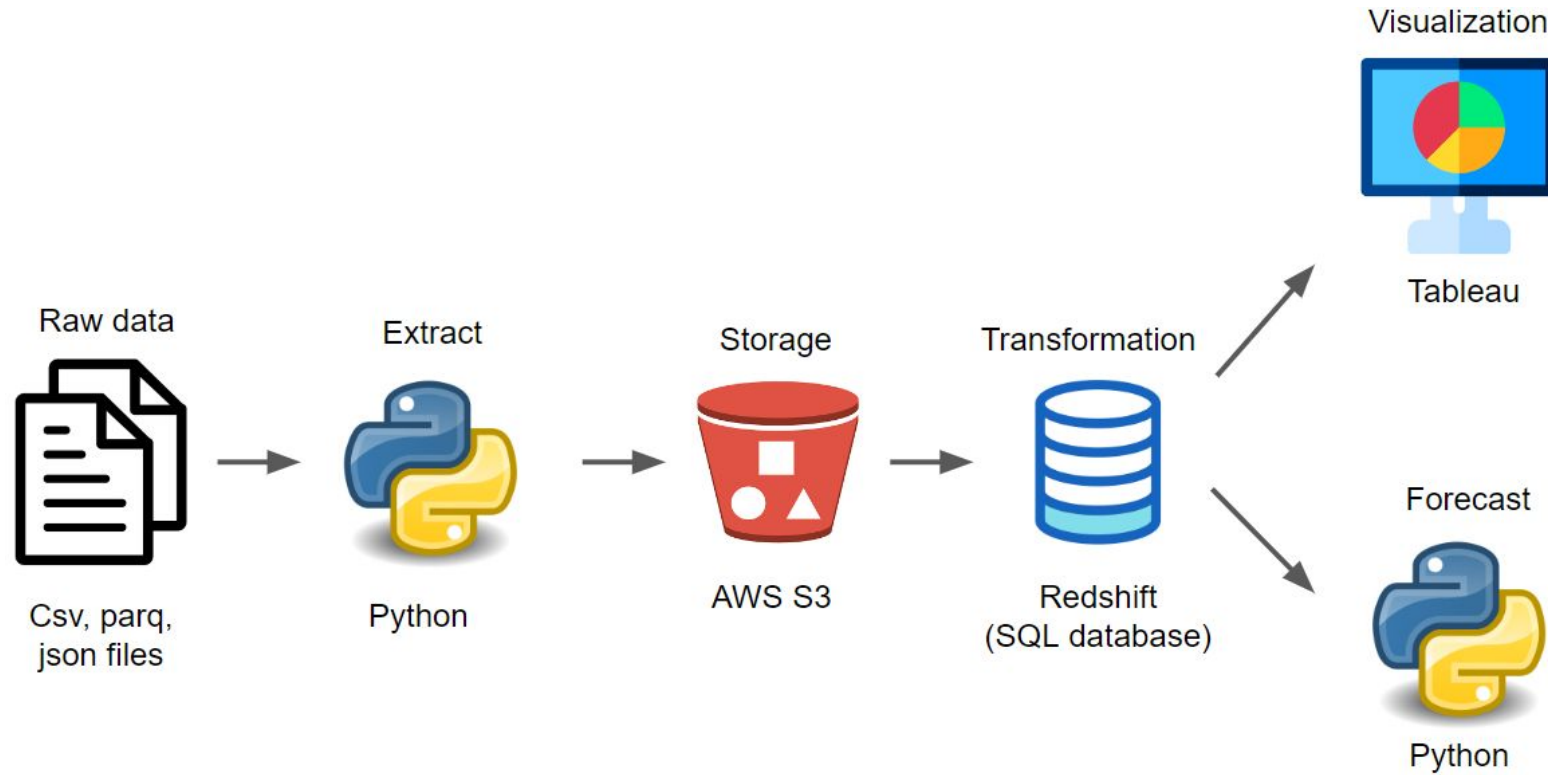
5. Conclusions

6. Q&A



2. DATA CLEANING

2.1 Data pipeline



2. DATA CLEANING

2.2 Data quality issues

a) Completeness

2 columns have some Null values:

- *product_category_name*: 53575 Null rows (3.75%)
- *EuroExchangeRate*: 99863 Null rows (6.99%)

b) Uniqueness

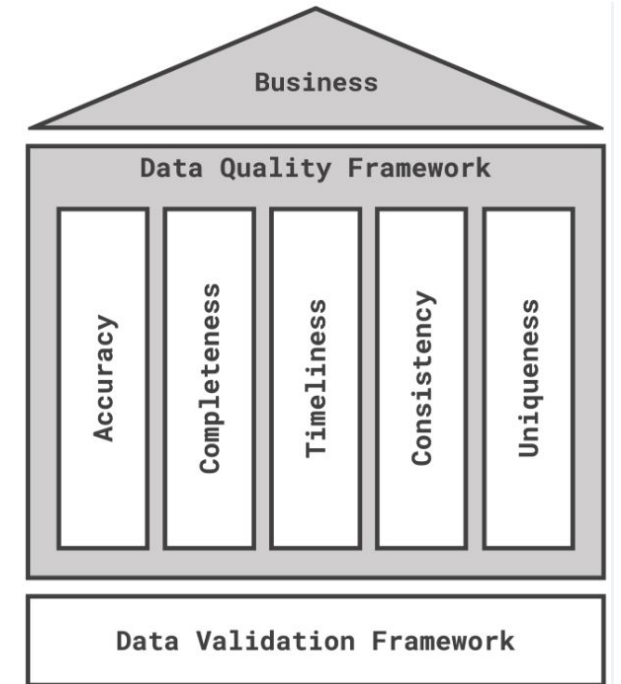
No issue

c) Consistency

No issue

d) Accuracy

- 55 records (0.004%) with negative values for quantity
- 2905 rows (0.2%) have questionable exchange rate for PLN and SEK currency:
 - Correct exchange rate is about 0.1-0.2
 - Exchange rate in our data is 1



2. DATA CLEANING

2.3 Data correction

- **Negative quantity** → change to 0 (low risk due to low number and cancelled order status)
- **Null product_category_name**: 3.75% → replace with product category of the same tenant + channel + product_id
 - result: 3.75% Null remain unchanged because there are no similar tenant + channel + product_id
- **Null Exchange rate**: 6.99% including
 - 6.64% have EUR currency → changed to 1 exchange rate → 0% Null
 - 0.35% have GBP currency → find exchange rate of GBP on same date, if not available then same month, if not available then using fixed value of 1.1 → 0% Null



OUTLINE

1. Introduction

2. Data cleaning

3. Data analysis

3.1 Performance over time

3.2 Channel

3.3 Product category

3.4 Order status

3.5 Price

4. Forecasting

5. Conclusions

6. Q&A

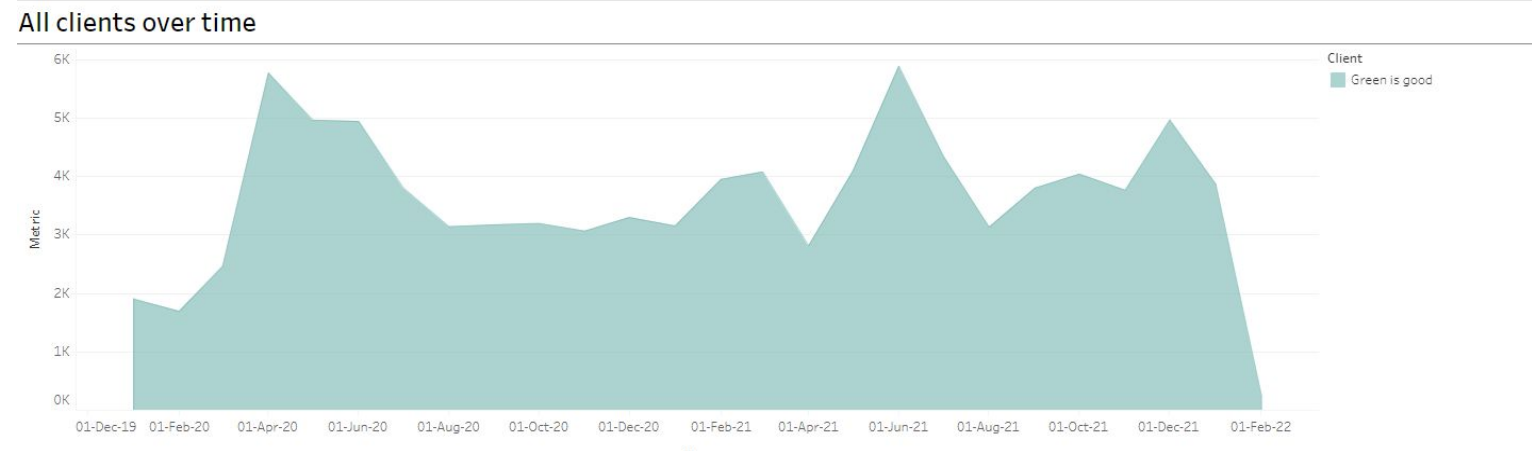
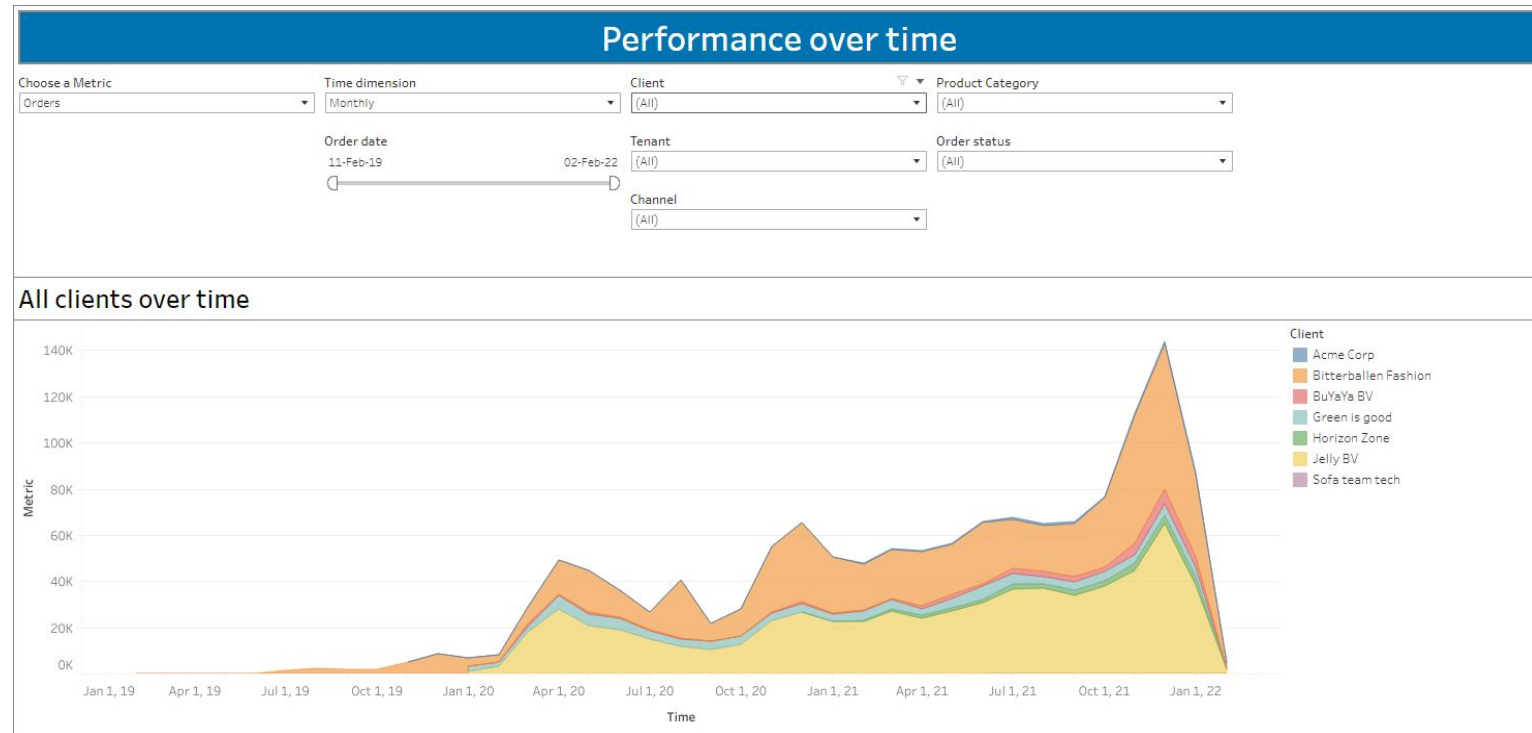


3. DATA ANALYSIS

[Visualization link](#)

3.1 Performance over time

- Total orders/quantity/revenue (across all clients) increase significantly from 2019 to 2022.
- **Bitterballen fashion** and **Jelly BV** are the 2 biggest clients
- **Green is good** has relatively stable performance and does not increase as quickly as other clients.



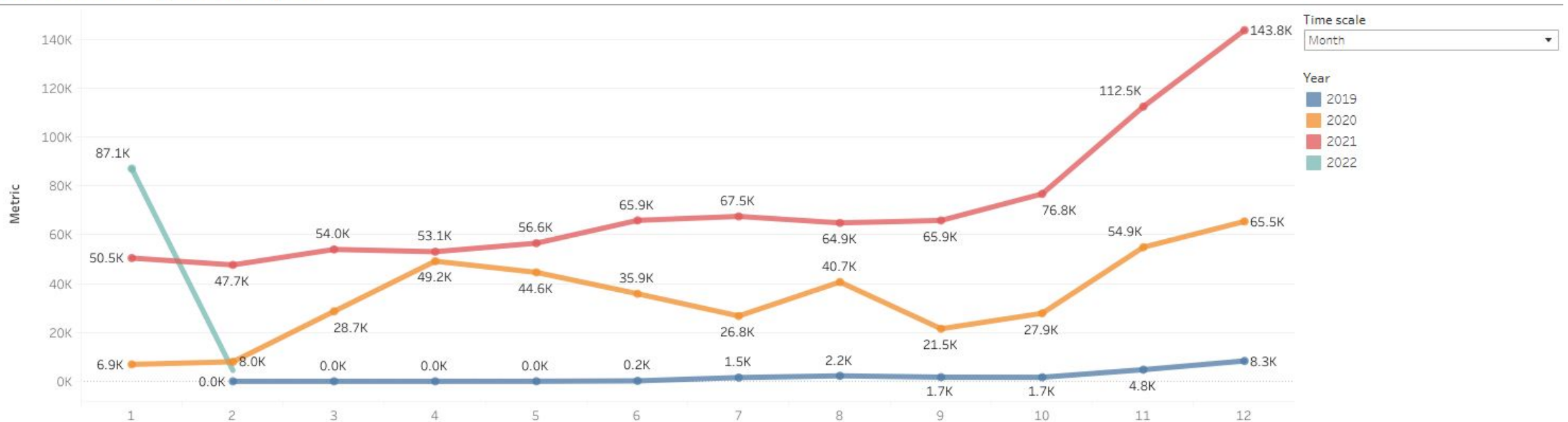
3. DATA ANALYSIS

[Visualization link](#)

3.1 Performance over time

- December has the best performance every year:
 - Seasonality due to Christmas/New Year holiday seasons
 - Overall increasing trend from January to December
- Spike in April 2020 probably related to corona lockdown
- February 2022 seems low because we only have order data until 2 Feb 2022.

Performance year over year



3. DATA ANALYSIS

[Visualization link](#)

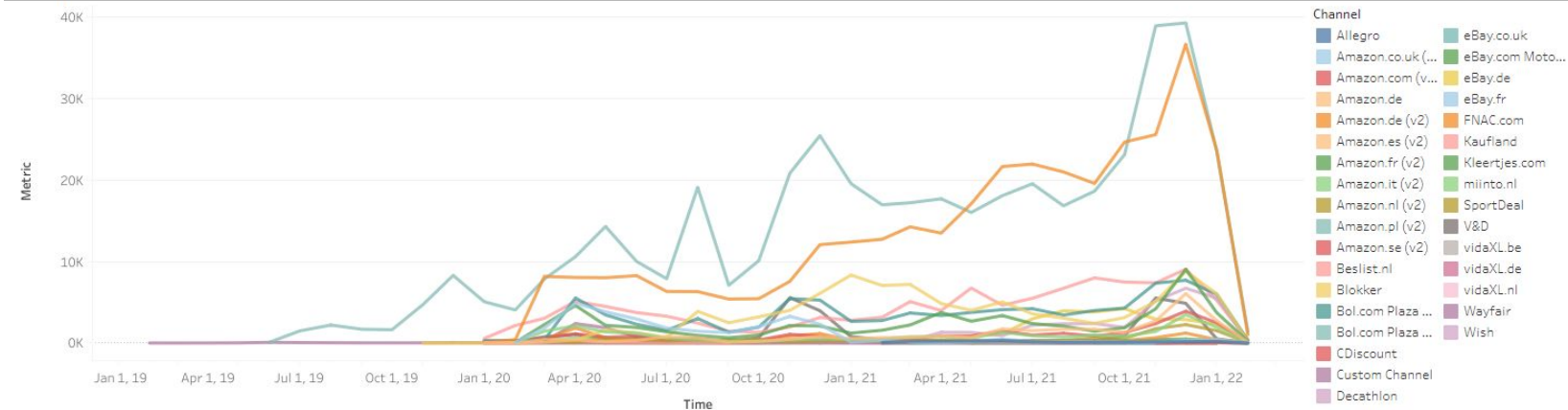
3.2 Channel

- Top 3 best performing channel overall is **Bol.com Plaza NL**, **Amazon.de**, and **Kaufland**.
- For each client and each tenant, the top 3 best-performing channel are different.
- Best-performing channels changed over time:
 - **Bol.com Plaza NL** was #1 for almost every month.
 - **Amazon.de** overtook to reach the top spot during summer 2021.
 - The 3rd spot was the competition between **Kaufland** and **Blokker**.

Best performing channels across all clients

Rank	Channel	
1	Bol.com Plaza NL (v3)	449.7K
2	Amazon.de (v2)	342.1K
3	Kaufland	108.9K

Compare channels over time



3. DATA ANALYSIS

[Visualization link](#)

3.3 Product category

- **Fietsaccessoires** is #1 in terms of orders & quantity.
- **Jongenfiets** is #1 in terms of revenue. This is because **Jongenfiets** has higher price than **Fietsaccessoires**.
- **Lagsnakers** is in top 3 for all metrics (orders, quantity, revenue).

Choose a Metric

Orders

Order date

11-Feb-19

02-Feb-22

Client

(All)

Product Category

(All)

Tenant

(All)

Order status

(All)

Channel

(All)

Best performing product categories across all clients

Rank	Product Category	
1	Fietsaccessoires	33.18K
2	Lage sneakers	28.58K
3	Verwarming	25.87K

Choose a Metric

Revenue (EUR)

Order date

11-Feb-19

02-Feb-22

Client

(All)

Product Category

(All)

Tenant

(All)

Order status

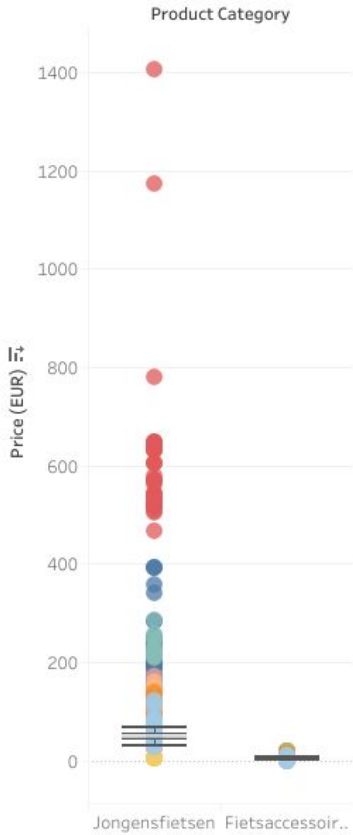
(All)

Channel

(All)

Best performing product categories across all clients

Rank	Product Category	
1	Jongensfietsen	1,343.97K
2	Lage sneakers	992.02K
3	Zwembad	808.18K

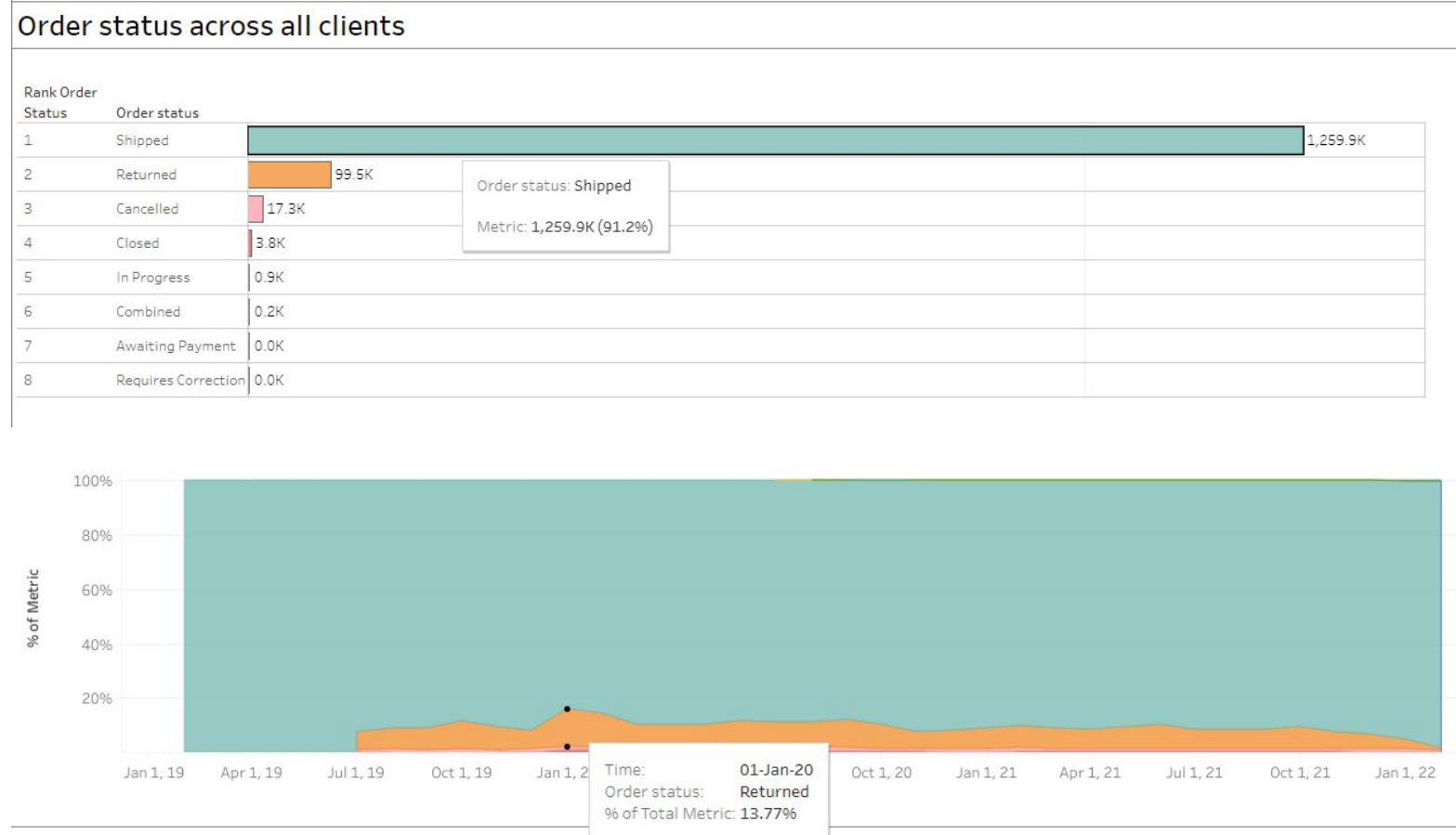


3. DATA ANALYSIS

[Visualization link](#)

3.4 Order status

- Distribution of order status:
 - Over 90% of orders have status **“Shipped”**
 - About 7% have status **“Returned”**.
 - About 1% of orders were **“Cancelled”**.
 - The other statuses account for very small percentage of orders.
- % of **“Returned”** orders decreased over time, from roughly 8% - 10% in 2019-2020 to about 7% in 2021.

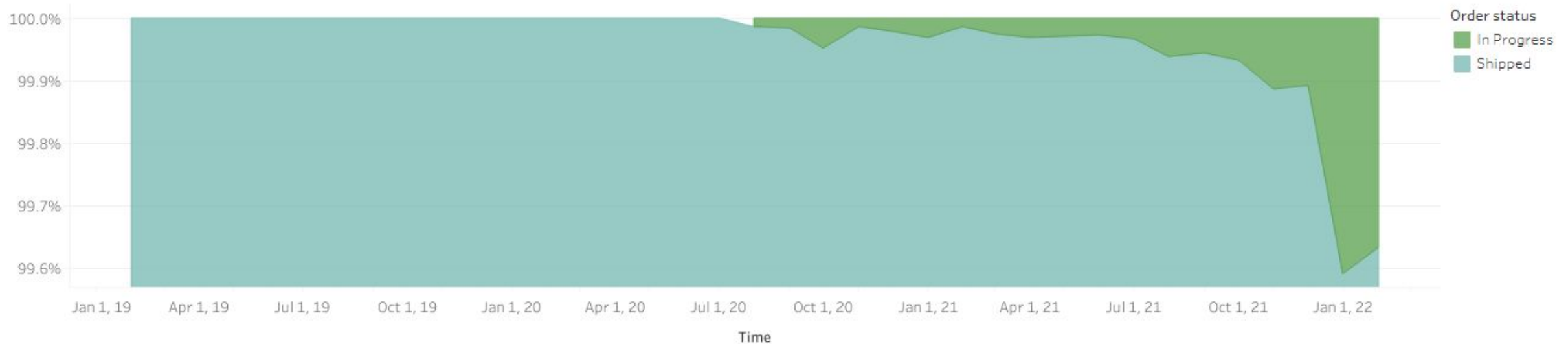


3. DATA ANALYSIS

[Visualization link](#)

3.4 Order status

- A large portion of orders in the latest months are still **“In progress”**.



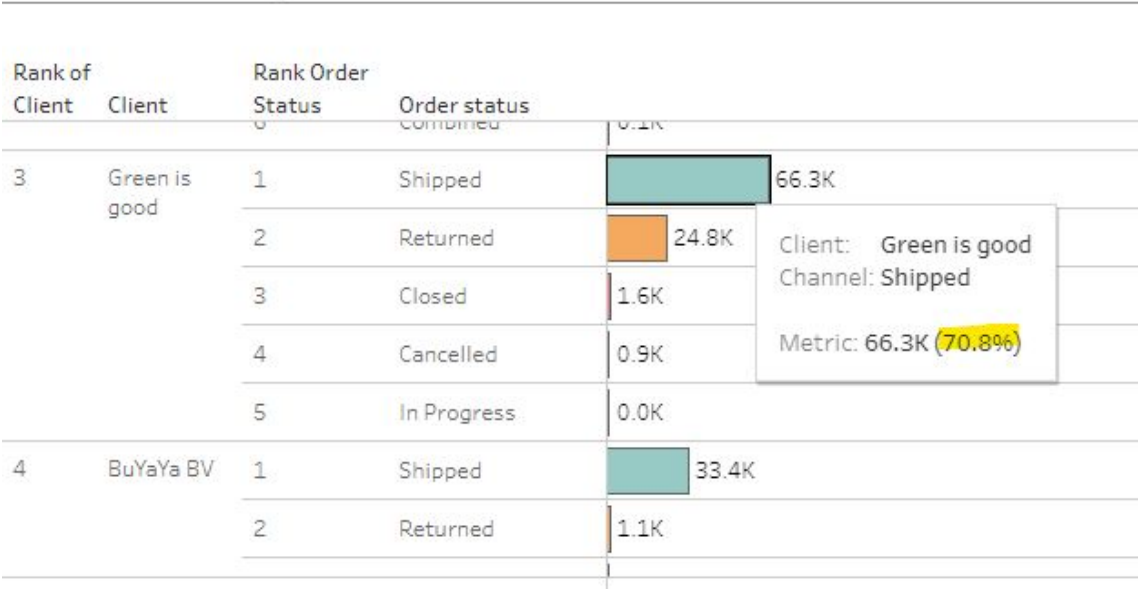
3. DATA ANALYSIS

[Visualization link](#)

3.4 Order status

- All clients have over 90% orders with status **Shipped**, except for **Green is good**.
- Only 70% of **Green is good**'s orders has status **Shipped**.
Returned orders make up 26.5% of all orders, which is quite alarming.

Order status per client

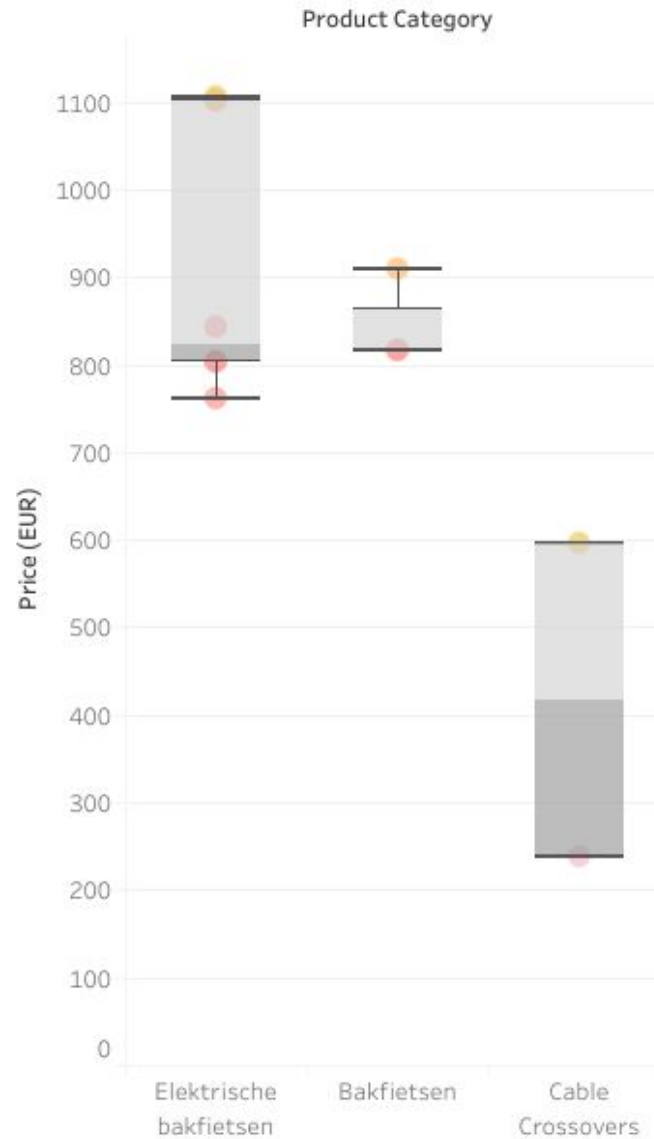


3. DATA ANALYSIS

[Visualization link](#)

3.5 Price

- Top 3 categories with highest median price are: **Elektrische bakfietsen**, **Bakfietsen** and **Cable crossovers**

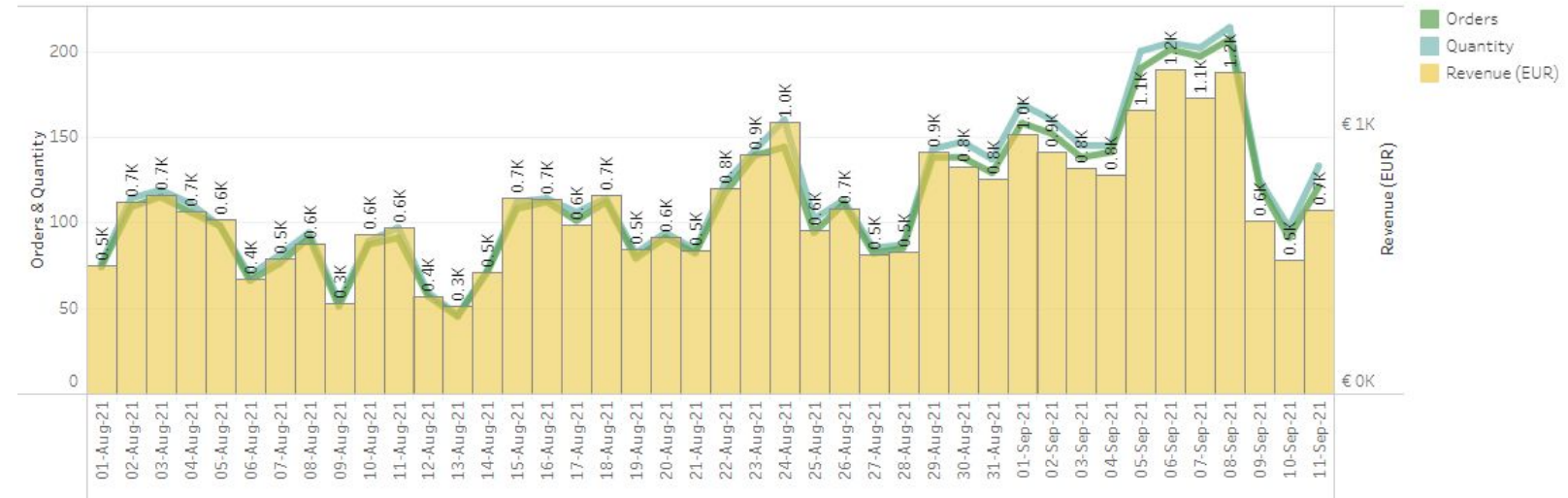
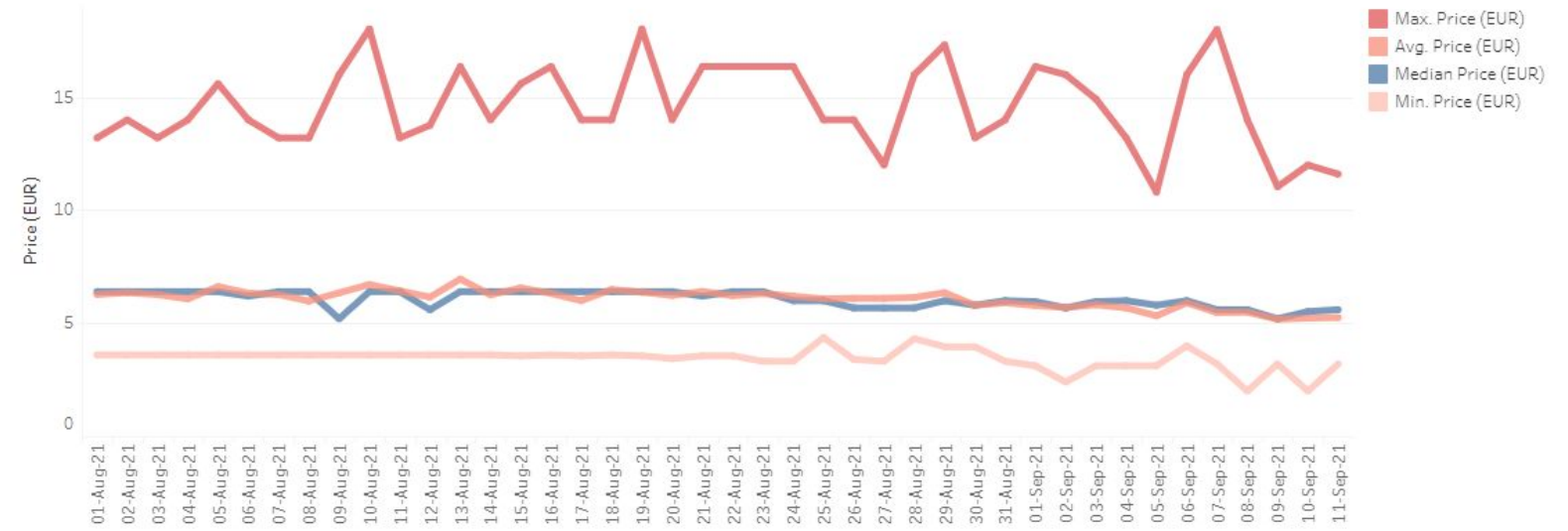


3. DATA ANALYSIS

[Visualization link](#)

3.5 Price

- We can monitor the change of price over time for each Product category, and study whether price is correlated with orders/quantity/revenue.
- It is better to limit the time-frame to a few weeks only because long-term trend is influenced by many other factors apart from price.
- Example: for **Fietsaccessoires** category, price decreased slightly in Aug-Sep 2021. Orders/Quantity/Revenue increased gradually over the same period of time.



OUTLINE

1. Introduction

2. Data cleaning

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4. Forecasting

4.1 Introduction to ARIMA

4.2 Model training

a) Non-seasonal ARIMA

b) Seasonal ARIMA

4.3 Validation

5. Conclusions

6. Q&A



4. FORECASTING

4.1 Introduction to ARIMA

- **AutoRegressive Integrated Moving Average (ARIMA)** is a popular model to forecast future value from historical time-series data. It includes 3 main parts:

- **I (Integrated)**: indicates the number of transformation that is done to make the time-series stationary.
- **AR (Autoregression)**: dependency of a particular record with a few past observations.

$$Y_t = \alpha + \beta_1 Y_{t-1} + \beta_2 Y_{t-2} + \dots + \beta_p Y_{t-p} + \epsilon_t$$

- **MA (Moving Average)**: dependency of a particular record with a few past errors.

$$Y_t = \alpha + \epsilon_t + \phi_1 \epsilon_{t-1} + \phi_2 \epsilon_{t-2} + \dots + \phi_q \epsilon_{t-q}$$

In summary:

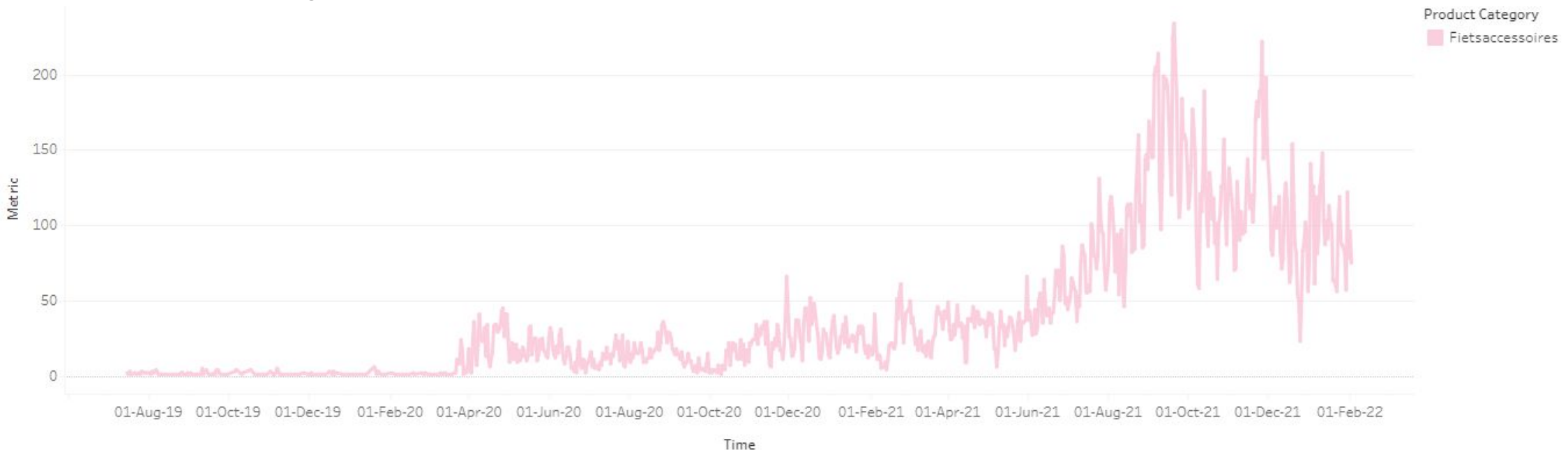
Predicted Y_t = Constant + Linear combination Lags of Y (up to p lags) + Linear Combination of Lagged forecast errors (up to q lags)

- A simple **(non-seasonal) ARIMA** model requires 3 parameters **(p, d, q)** for training, which correspond to 3 component of ARIMA: AR, I and MA.
- A **seasonal ARIMA (SARIMA)** model can be used to train a seasonal time-series. In addition to **(p, d, q)**, SARIMA requires some extra parameters **(P, D, Q, M)**. **M** is the number of period in a cycle (eg. 12 for monthly data, 4 for quarterly data). **(P, D, Q)** is similar to **(p, d, q)**. The only difference is that **(P, D, Q)** come from the seasonality instead of the original time series.

4. FORECASTING

4.1 Introduction to ARIMA

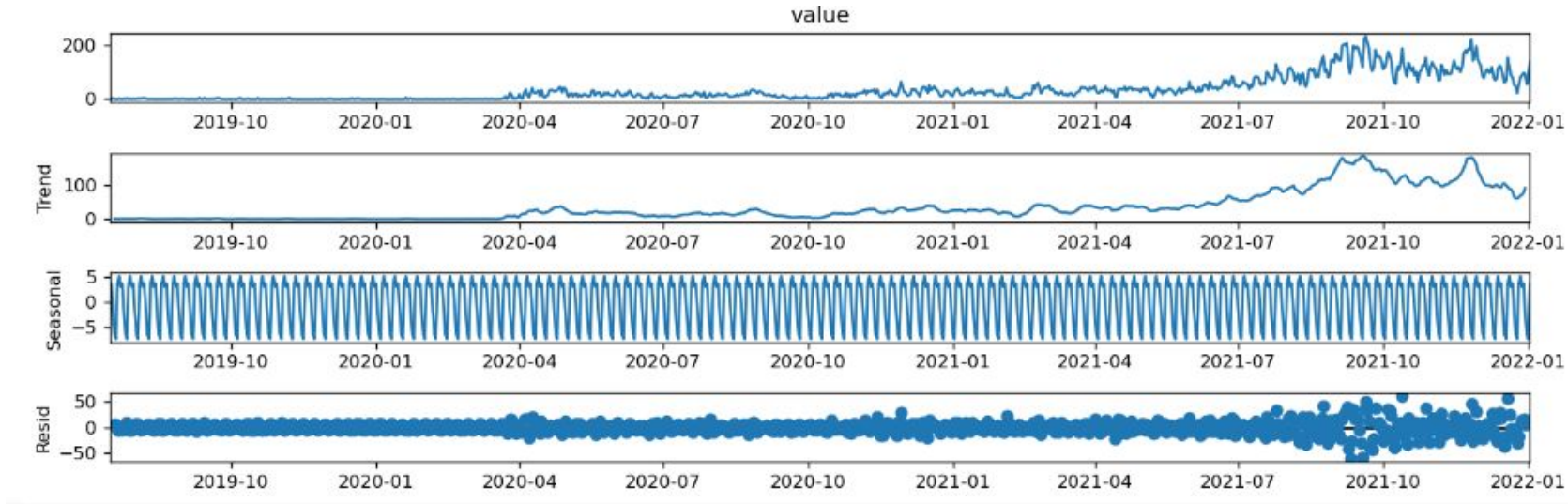
- **Data selected for this project:** daily quantity data of Product category **Fietsaccessoires**
 - highest quantity among the categories
 - sold by 1 tenant only: Bitterballen BV
 - low % of cancellation, so excluding cancelled orders will not affect the overall quantity a lot
 - data ranges from 16 Jul 2019 to 2 Feb 2022:
 - Training data is from 16 Jul 2019 to 2 Jan 2022.
 - Testing data is from 3 Jan 2022 to 2 Feb 2022.



4. FORECASTING

4.1 Introduction to ARIMA

Decomposition of our time series:



There is seasonality, so seasonal ARIMA should be studied. We will compare 4 different ARIMA models:

- Non-seasonal ARIMA by manual setting
- Non-seasonal ARIMA by automatic search
- Seasonal ARIMA by manual setting
- Seasonal ARIMA by automatic search

4. FORECASTING

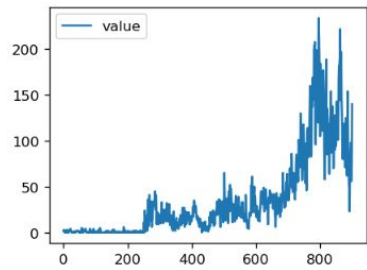
4.2 Data training a) Non-seasonal ARIMA

Manual setting of (p, d, q)

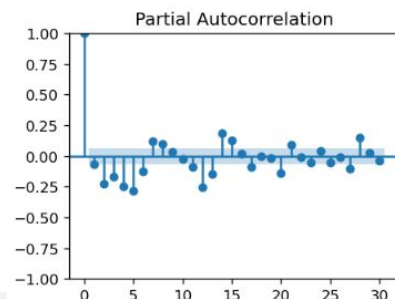
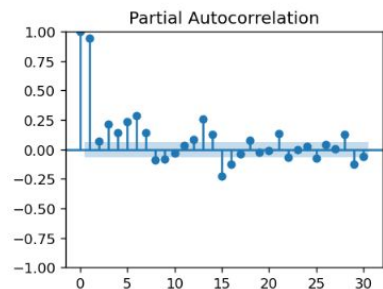
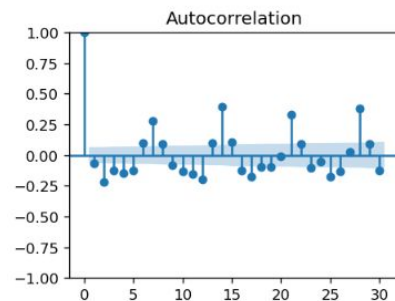
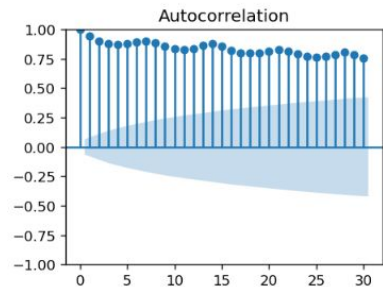
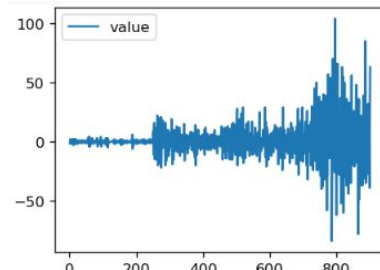
1st differencing makes the series stationary $\rightarrow d = 1$

No lags have clear correlation in PACF and ACF of 1st differencing (except for lag 7, 14, 21,... due to seasonality) $\rightarrow$ set $p = 0$ and $q = 0$

Original series



1st differencing



```
# Augmented Dickey-Fuller test (if p < 0.05 --> the series is stationary)
adf_test = adfuller(df_train)
print(f'p-value original series: {adf_test[1]}')

adf_test = adfuller(df_train.diff().dropna())
print(f'p-value 1st differencing: {adf_test[1]}')

adf_test = adfuller(df_train.diff().diff().dropna())
print(f'p-value 2nd differencing: {adf_test[1]}')

# from the above result, 1st differencing is stationary --> set d = 1
# Looking at the ACF & PACF graph of 1st differencing, correlation of all lags are week --> set p = 0 and q = 0

p-value original series: 0.6824788745731261
p-value 1st differencing: 1.4016618701845823e-11
p-value 2nd differencing: 1.4830192174967184e-22
```

4. FORECASTING

4.2 Data training a) Non-seasonal ARIMA

Auto search of (p, d, q)

Auto search gives (p, d, q) of (5, 1, 2).

Manual and auto approach have the same $d = 1$, so we can compare fitting metrics.

Parameters from Auto search give a model that fit our training data better:

- Log Likelihood of auto search is less negative
- AIC, BIC, HQIC of auto search are smaller

Manual with (0, 1, 0)

Log Likelihood	-3702.994
AIC	7407.987
BIC	7412.791
HQIC	7409.822

Auto with (5, 1, 2)

Log Likelihood	-3534.912
AIC	7085.824
BIC	7124.252
HQIC	7100.503

4. FORECASTING

4.2 Data training b) Seasonal ARIMA

Manual setting of (P, D, Q)

We will use the same (p, d, q) as our manual approach (0, 1, 0)

Set M = 7 (seasonality every 7 lags)

The original seasonality series is already stationary, no need to differencing → Set D = 0

From PACF and ACF, we determine P = 1 and Q = 4

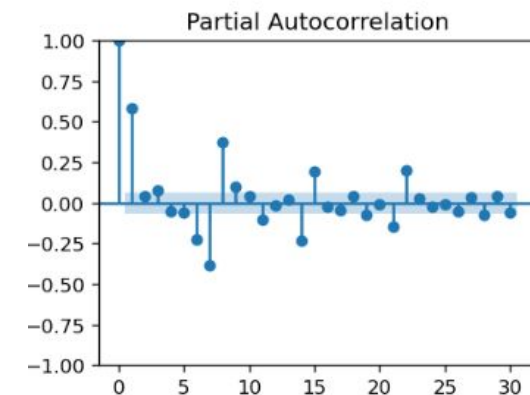
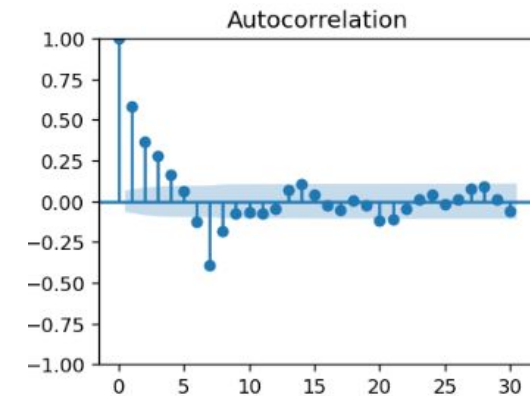
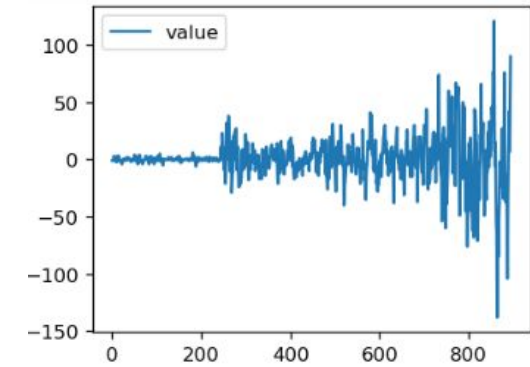
```
# Augmented Dickey-Fuller test
adf_test = adfuller(df_seasonal)
print(f'p-value original seasonality: {adf_test[1]}')

adf_test = adfuller(df_seasonal.diff().dropna())
print(f'p-value seasonality 1st differencing: {adf_test[1]}')

adf_test = adfuller(df_seasonal.diff().diff().dropna())
print(f'p-value seasonality 2nd differencing: {adf_test[1]}')
```

```
# Original seasonality is already stationary --> set D = 0
# Lag 1 in PACF graph is significant --> set P = 1
# Lag 1 to 4 in ACF graph is significant --> set Q = 4
# Cycle happens every 7 days --> set M = 7
```

```
p-value original seasonality: 7.114943259504384e-08
p-value seasonality 1st differencing: 3.2087504162783935e-21
p-value seasonality 2nd differencing: 3.1158536909626216e-23
```



4. FORECASTING

4.2 Data training b) Seasonal ARIMA

Manual setting of (P, D, Q)

p-value of 3rd and 4th coefficients of Q are non-significant → try reducing Q from 4 to 2

result: no significant improvement in fitting

Manual with (0, 1, 0) (1, 0, 4, 7)



Manual with (0, 1, 0) (1, 0, 2, 7)

Log Likelihood	-3554.240
AIC	7120.480
BIC	7149.301
HQIC	7131.489

Log Likelihood	-3554.435
AIC	7116.870
BIC	7136.084
HQIC	7124.210

4. FORECASTING

4.2 Data training b) Seasonal ARIMA

Auto search of (p, d, q) (P, D, Q, M)

Set $d = 1$, $D = 0$ and $M = 7$

Auto search gives (p, d, q) (P, D, Q, M) of (5, 1, 0) (2, 0, 3, 7)

Again, parameters from Auto search give a model that fit our training data better:

- Log Likelihood of auto search is less negative
- AIC, BIC, HQIC of auto search are smaller

Manual with (0, 1, 0) (1, 0, 2, 7)

Log Likelihood	-3554.435
AIC	7116.870
BIC	7136.084
HQIC	7124.210

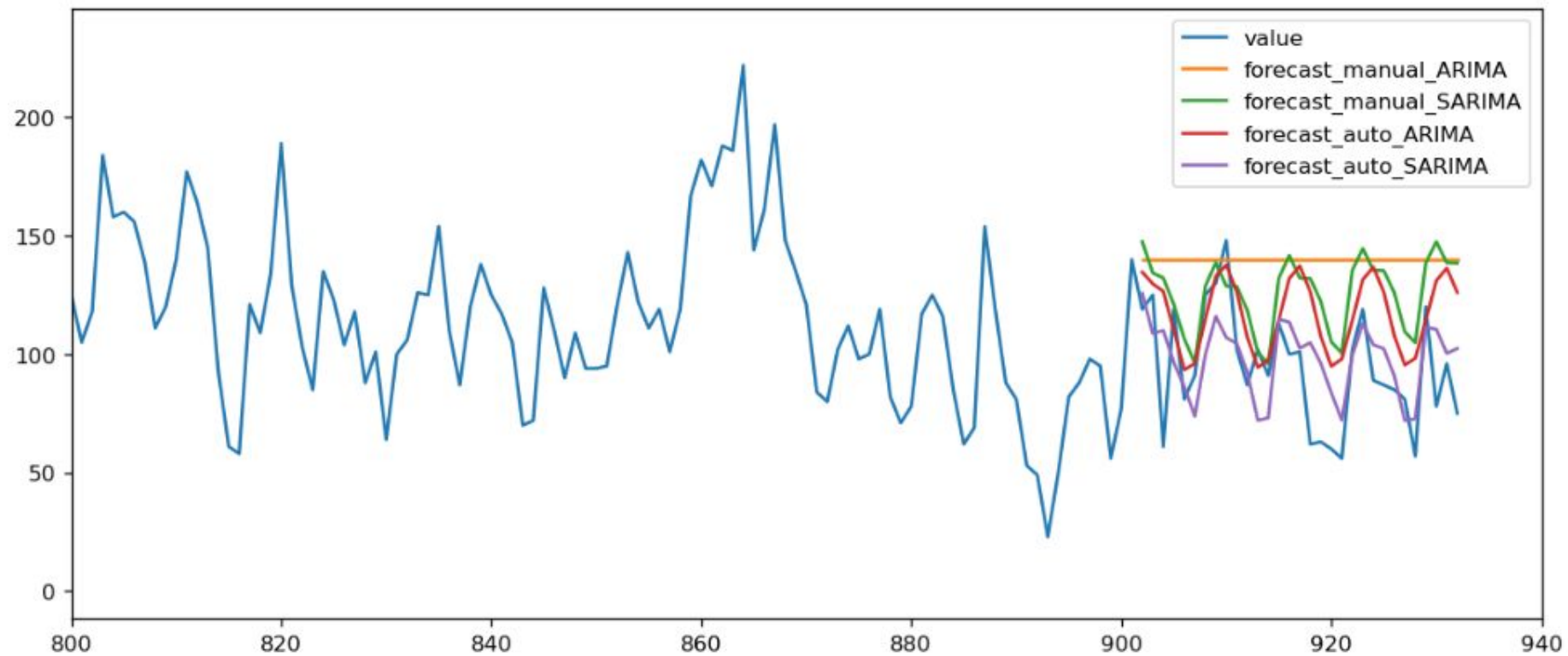
Auto with (5, 1, 0) (2, 0, 3, 7)

Log Likelihood	-3492.086
AIC	7006.171
BIC	7059.010
HQIC	7026.355

4. FORECASTING

4.1 Validation

- Plotting forecast from 4 different models against the actual data:
 - Except for Auto SARIMA, the other 3 models over-forecast the quantity because there is a downward trend after December 2021, which is difficult to catch without January data in the training set.



4. FORECASTING

4.1 Validation

- Seasonal model performs better than the corresponding non-seasonal model
- Auto search improves the performance significantly
- From best to worst: Auto seasonal >>> Auto non-seasonal >>> Manual seasonal >>> Manual non-seasonal

Manual with (0, 1, 0)

```
mae - manual - ARIMA: 46.16129032258065  
mape - manual - ARIMA: 0.5924738712629284  
rmse - manual - ARIMA: 51.47219731830179
```

Manual with (0, 1, 0) (1, 0, 2, 7)

```
mae - manual - SARIMA: 32.71230388197109  
mape - manual - SARIMA: 0.4171171195489108  
rmse - manual - SARIMA: 38.70189361481308
```

Auto with (5, 1, 2)

```
mae - auto - ARIMA: 25.50872487347351  
mape - auto - ARIMA: 0.33238012853962373  
rmse - auto - ARIMA: 31.75083112002245
```

Auto with (5, 1, 0) (2, 0, 3, 7)

```
mae - auto - SARIMA: 17.038138595194035  
mape - auto - SARIMA: 0.20780635484351895  
rmse - auto - SARIMA: 21.273850733236483
```

5. CONCLUSIONS

Data analysis

- Performance improves over time for all clients, except for **Green is good**.
- December is the best-performing month in a year.
- **Bol.com Plaza NL** is the best-performing channel overall, whereas **Amazon.de** is #2. The #3 position is different each month.
- **Fietsaccessories** is the best-performing category in terms of orders and quantity, while **Jongenfiets** is in the top spot for revenue. This is reasonable because **Jongenfiets** has higher price than **Fietsaccessories**.
- Over 90% of orders are shipped, 7% are returned and 1% are cancelled.
 - % returned orders decreased gradually over the observed period.
 - **Green is good** has very high % of returned orders (nearly 30%) compared to other clients (less than 10%) → to check the quality of their products, customer service, ordering and shipping process.
- Top 3 categories with highest median price are **Elektrische bakfietsen**, **Bakfietsen** and **Cable crossovers**.
- Checking Price and Orders/Quantity/Revenue over the short period of time can reveal if there is any correlation between Price and Performance.

5. CONCLUSIONS

Forecasting

- **ARIMA** was employed to predict future values for time series data.
- Seasonal models fit the training data better, and auto search outperforms manual approach.
- **Seasonal ARIMA by auto search** gave the best result when validating against the testing set of Product Category “**Fietsaccessoires**”.
- It is difficult to achieve low error because our data has a strong spike in December 2021. Our last date of training data is 2 January, so it was hard to capture the decreasing trend after December.
- Train on weekly data:
 - Reduced noise compared daily data
- Monthly data is not a good choice:
 - Limiting data (only 3 years of data → fewer than 40 observations)
 - Some product categories have fewer than 12 observations
- Potential future works:
 - Benchmark ARIMA against other forecasting models (eg. Linear regression, xgboost, etc)
 - Add more exogeneous variables into the ARIMA model to capture holiday effect
 - For low-volume product categories, we can train models on all tenants, then apply the coefficients to each tenant. Another solution is to train model on similar product categories which have higher order volume.



Questions?